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# The aciclovir metabolite CMMG is detectable in the CSF of subjects with neuropsychiatric symptoms during aciclovir and valaciclovir treatment

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***Objectives*: Neuropsychiatric symptoms related to aciclovir or valaciclovir treatment have been a problem since aciclovir was introduced in the early 1980s. We have previously found that subjects with aciclovir-related neuropsychiatric symptoms have increased serum concentrations of aciclovir's main metabolite, 9-carboxymethoxymethylguanine (CMMG). The aim of this study was to investigate whether CMMG was present in the CSF of aciclovir- or valaciclovir-treated subjects with or without neuropsychiatric side effects that appeared during therapy.**

***Methods*: We investigated retrospectively CSF collected from 21 aciclovir- or valaciclovir-treated subjects. Of these, 9 were subjects with neuropsychiatric signs and symptoms and 12 were asymptomatic subjects, including 10 subjects from a valaciclovir multiple sclerosis trial and 2 subjects with recurrent herpes encephalitis.**

***Results*: CMMG could only be detected in the CSF of subjects with neuropsychiatric symptoms and signs (median CMMG concentration 1.0 μmol/L, range 0.6–7.0). The concentration of CMMG was below the limit of quantification (<0.5 μmol/L) in asymptomatic subjects ($P < 0.001$). All patients with neuropsychiatric signs and symptoms, except one, had acute renal function impairment or chronic renal failure.**

***Conclusions*: These results are consistent with the hypothesis that CMMG is involved in the development of neuropsychiatric side effects in aciclovir- or valaciclovir-treated patients. Measurement of CMMG in CSF and/or serum is a promising tool in the diagnostic procedure for aciclovir- or valaciclovir-treated patients with neuropsychiatric symptoms and may help to differentiate between side effects and herpes encephalitis.**

Keywords: serious adverse reactions, 9-carboxymethoxymethylguanine, neurotoxicity, renal failure

## Introduction

Neuropsychiatric side effects related to aciclovir treatment were first reported in the early 1980s.[1] Symptoms such as tremor, visual and auditory hallucinations, confusion and coma have been described.[2] Recently, patients treated using valaciclovir (the L-valyl ester of aciclovir with improved bioavailability compared with oral aciclovir), have been reported to develop neuropsychiatric signs and symptoms.[3,4] Predisposing factors are acute or chronic renal failure, simultaneous treatments with other neurotoxic drugs and severe illness. However, there are case reports describing CNS symptoms in subjects without renal failure.[5–7] The aetiology of the neuropsychiatric side effects is unknown. The results of earlier studies of the relationship between CNS symptoms and serum/plasma concentrations of aciclovir are inconsistent.[8–10] There is a need for a simple test to distinguish between neurological symptoms caused by encephalitis and neuropsychiatric symptoms caused by the aciclovir or valaciclovir treatment.

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Renal excretion of unchanged drug is the major route of aciclovir elimination in healthy individuals. In renal failure subjects, a considerable proportion of aciclovir is metabolized to 9-carboxymethoxymethylguanine (CMMG), probably through the action of alcohol dehydrogenase (ADH) and aldehyde dehydrogenase (ALDH). A minor metabolite, 8-OH-aciclovir, is also formed, but to a lesser extent (<0.2%) (Figure 1).[11–13] We have recently shown that serum concentrations of CMMG are consistently increased in subjects with aciclovir-induced neuropsychiatric symptoms.[14] The aim of this study was to investigate whether detectable concentrations of CMMG are present in CSF during treatment with aciclovir or valaciclovir in patients with or without neuropsychiatric symptoms.

## Methods

### *Subjects*

A total of 21 subjects were studied (Table 1), 12 without and 9 with neuropsychiatric symptoms. Of the asymptomatic subjects, 10 had multiple sclerosis (MS) and 2 had relapsing herpes encephalitis. The subjects with MS participated in a pharmacokinetic study of valaciclovir[15] which was part of a larger investigation in which the possible involvement of herpes viruses in the pathogenesis of MS was explored.[16] Nine subjects had neuropsychiatric symptoms and were treated by either intravenous or oral administration of aciclovir or valaciclovir. The indications for treatment are presented in Table 1. In the symptomatic group, four subjects had chronic renal failure and one patient had normal renal function.[7] Four patients developed acute renal function impairment during treatment. Two patients in the symptomatic group received doses that were too high in relation to their renal function, one received a higher valaciclovir dose than recommended (6 g) but had no initial renal impairment, and six had doses adjusted to their renal function according to standard recommendations. Subject characteristics are summarized in Table 1. The neuropsychiatric symptoms of the nine symptomatic subjects were reviewed to exclude other possible causes.

The Regional Ethics Committee of Karolinska Institutet approved the study (no. 379/98). Subjects included in the MS study by Lycke, Malmeström and Ståhle gave written informed consent.[15] In agreement with the ethics committee in 1998, written informed consent was not regarded necessary for the remaining patients since this is a retrospective study and no sample was taken for research purposes. The subjects were treated at the Departments of Infectious Diseases, Neurology, Haematology or Nephrology in Swedish hospitals and at the Department of Internal Medicine, Soroka Medical Center, Beer Sheva, Israel.

The aciclovir and CMMG results from the serum samples of subjects included in this study have been described previously,[14] except for those of subjects 20 and 21. A case report on aciclovir-related signs and symptoms has also been published.[7]

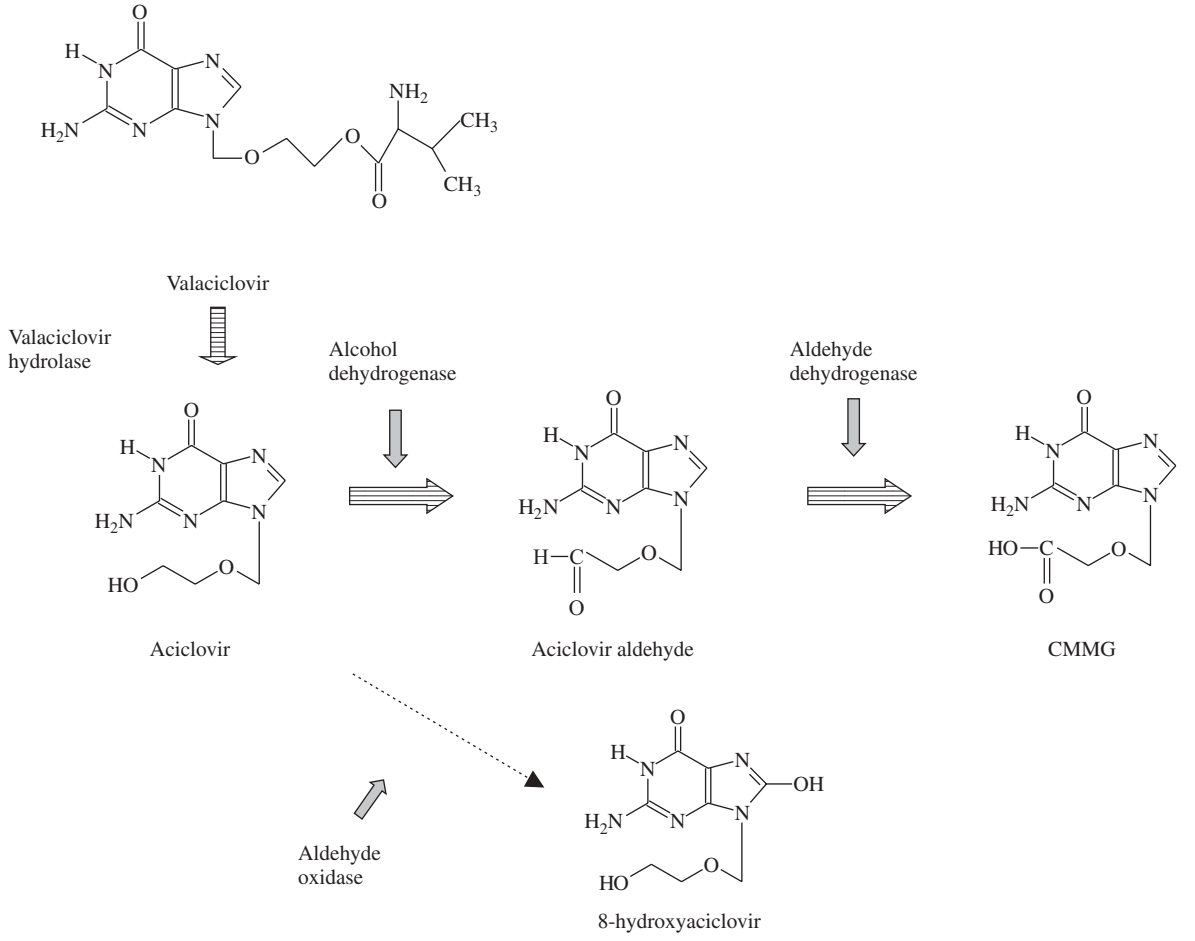



**Figure 1.** Metabolic pathways proposed for valaciclovir and aciclovir. According to previous studies the formation of 8-hydroxyaciclovir in human subjects is <0.2%. (CMMG = 9-carboxymethoxymethylguanine).



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**Table 1.** Characteristics of subjects without (1–12) and with (13–21) neuropsychiatric side effects, their treatment and aciclovir (ACV) and 9-carboxymethoxymethylguanine (CMMG) serum concentrations

| Subject | Age (years) | Sex | Weight (kg) | Estimated $CL_{CR}$ (mL/min) | Indication for ACV treatment | Route | Dose/day (mg) | Lag time from dose until serum sample (h) | Serum concentration (μmol/L) | | Symptoms |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | ACV | CMMG | |
| 1 | 38 | F | 90 | 112 | multiple sclerosis | oral | 3000 VCV | 2 + 8 | 9.0[a] | 0.7[a] | no |
| 2 | 20 | F | 62 | 91 | multiple sclerosis | oral | 3000 VCV | 2 + 8 | 7.3[a] | 1.0[a] | no |
| 3 | 31 | M | 85 | 109 | multiple sclerosis | oral | 3000 VCV | 2 + 8 | 6.7[a] | 0.6[a] | no |
| 4 | 36 | F | 62 | 68 | multiple sclerosis | oral | 3000 VCV | 2 + 8 | 14.6[a] | 1.0[a] | no |
| 5 | 25 | F | 64 | 85 | multiple sclerosis | oral | 3000 VCV | 2 + 8 | 7.5[a] | 0.4[a] | no |
| 6 | 25 | M | 87 | 130 | multiple sclerosis | oral | 3000 VCV | 2 + 8 | 8.4[a] | 0.6[a] | no |
| 7 | 33 | F | 70 | 89 | multiple sclerosis | oral | 3000 VCV | 2 + 8 | 8.2[a] | 0.8[a] | no |
| 8 | 32 | F | 53 | 65 | multiple sclerosis | oral | 3000 VCV | 2 + 8 | 8.6[a] | 0.9[a] | no |
| 9 | 25 | F | 57 | 84 | multiple sclerosis | oral | 3000 VCV | 2 + 8 | 5.9[a] | 0.7[a] | no |
| 10 | 46 | F | 60 | 69 | multiple sclerosis | oral | 3000 VCV | 2 + 8 | 7.4[a] | 0.8[a] | no |
| 11 | 71 | M | 54 | 59 | relapsing herpes encephalitis | oral | 2000 VCV | 13 | 3.5 | 0.9 | no |
| 12 | 40 | F | NA | NA | relapsing herpes encephalitis | oral | 4000 ACV | 4 | 3.9 | 0.9 | no |
| 13 | 28 | M | NA | 8 | prophylaxis after transplantation | iv | 250 ACV | 25 | 1.4 | 10.1 | yes |
| 14 | 47 | M | 70 | 11 | CMV-infection | oral | 2400 ACV | 48 | 22.5 | 161.0 | yes |
| 15 | 65 | M | 54 | 5 | varicella zoster | oral | 1600 ACV | 16 | 22.7 | 35.1 | yes |
| 16 | 68 | M | 138 | 21 | suspected encephalitis | iv | 3750 ACV | 6 | 97.9 | 56.3 | yes |
| 17 | 68 | M | 57 | 9 | varicella zoster | oral | 3200 ACV | 6 | 8.7 | 18.0 | yes |
| 18 | 52 | M | 69 | 123 | varicella zoster | oral | 2000 ACV | NA | NA | NA | yes |
| 19 | 63 | F | 62 | 18 | herpes mucositis | iv | 500 ACV | 72 | 6.0 | 6.0 | yes |
| 20 | 59 | F | 73 | 32 | relapsing herpes encephalitis | oral | 6000 VCV | <24 | 81.8 | 26.3 | yes |
| 21 | 35 | M | 60 | 19 | herpes mucositis | iv | 500 ACV | 16 | 38.8 | 28.9 | yes |

ACV= oral aciclovir; $CL_{CR}$ = estimated creatinine clearance in mL/min according to calculation using the Cockcroft–Gault formula; CMMG= 9-carboxymethoxymethylguanine; CMV= cytomegalovirus; iv= intravenous; NA= not available; VCV= valaciclovir.
[a]Geometric mean of samples collected at 2 and 8 h after the dose.

The CSF sample from this subject was stored frozen and analysed after the case was published.

### *Samples*

The CSF samples were obtained from MS subjects at steady state 2 and 8 h after oral administration of 1000 mg of valaciclovir, except for one subject from whom a single CSF sample was obtained at 2 h after the valaciclovir dose and another subject from whom a single CSF sample was obtained at 8 h after the valaciclovir dose. The CSF aciclovir concentrations of the 10 MS subjects are given as geometric mean values of the two samples. CSF samples from the remaining two subjects in the asymptomatic group were collected 4 and 13 h, respectively, after the administration of the dose. Subjects 16, 17, 18 and 21 in the symptomatic group had a lag time of 6–8 h between the dose administration and collection of the CSF sample. For subjects 13, 14, 15 and 20 the lag time could be estimated from the patient charts to be <24, <8, <12 and <24 h, respectively. Sample time for subject 19 was unknown. The reason for these variations in sampling time is that samples from the subjects with neuropsychiatric symptoms were obtained during diagnostic procedures.

Serum samples were collected from all subjects except one. The lag times from the last dose to the sample are given in Table 1. CSF and serum samples were stored at –20°C until processing. Aciclovir and CMMG concentrations were determined at the Department of Clinical Pharmacology, Karolinska University Hospital, Huddinge, Stockholm, Sweden, using solid phase extraction followed by high-performance liquid chromatography with fluorescence detection.[17] The limit of detection (LOD) was 0.2 μmol/L and the limit of quantification (LOQ) was 0.5 μmol/L.

### *Neuropsychiatric symptoms*

The type and frequency of signs and symptoms in subjects with aciclovir-induced side effects were compiled and compared with earlier findings.[1–10,14]

### *Statistics*

CSF concentrations of aciclovir and CMMG are given as median and range. Owing to the pharmacokinetic differences between subjects with and without renal failure, plasma concentrations of aciclovir in the latter group were taken as the geometric mean of the concentrations at 2 and 8 h, which is a crude but reasonable estimate of exposure. Estimated $CL_{CR}$, calculated using the Cockcroft–Gault formula,[18] is given as mean ± SD. The Mann–Whitney *U*-test was used to test the hypothesis since many concentrations of CMMG were below the LOQ.



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## Results

Measurable CMMG concentrations in CSF were found only in subjects with neuropsychiatric symptoms. The median CMMG concentration in this group was 1.0 μmol/L (range 0.6–7), whereas all asymptomatic subjects had CMMG concentrations below the LOD ($P < 0.001$; Figure 2). Aciclovir concentrations in the CSF were significantly higher in subjects with neuropsychiatric signs and symptoms (median 8.6 μmol/L, range 3.5–28.6) than in asymptomatic subjects (median 2.4 μmol/L, range 1.1–5.1) ($P < 0.001$; Figure 2). All symptomatic subjects except one had severe renal failure with a mean estimated creatinine clearance of 15 ± 9 mL/min. Asymptomatic subjects had a mean estimated creatinine clearance of 87 ± 22 mL/min (Table 1).

The median concentration of CMMG in serum was 27.6 μmol/L (range 6.0–161.0) in symptomatic subjects and 0.8 μmol/L (range 0.5–1.0) ($P < 0.001$) in asymptomatic subjects. The corresponding median concentration of aciclovir in serum was 22.6 μmol/L (range 1.4–97.9) in symptomatic subjects and 8.1 μmol/L (range 3.5–14.6) in asymptomatic subjects ($P = 0.1$).

No factors other than treatment with aciclovir or valaciclovir could be identified as the cause of the neuropsychiatric signs and symptoms.

### *Neuropsychiatric symptoms*

Eight patients had no initial CNS signs or symptoms, except that one patient with relapsing herpes encephalitis had herpes-related headache. They developed neuropsychiatric symptoms after aciclovir treatment was commenced or after the dose was increased. Symptoms were confusion, somnolence, hyperreflexia, myoclonus, hallucinations, incoherence and unresponsiveness. All symptoms were similar to those described in earlier publications.[1–10,14] The ninth patient had signs and symptoms of an acute CNS infection before aciclovir treatment was initiated. The patient presented with confusion and aphasia, and herpes encephalitis was suspected. Aciclovir treatment was initiated and his status improved. Six days later, confusion, hallucinations, somnolence, ataxia and slurred speech developed in close relation to acute deterioration of renal function. The condition improved only after cessation of aciclovir therapy. Encephalitis could not be confirmed despite thorough investigations.

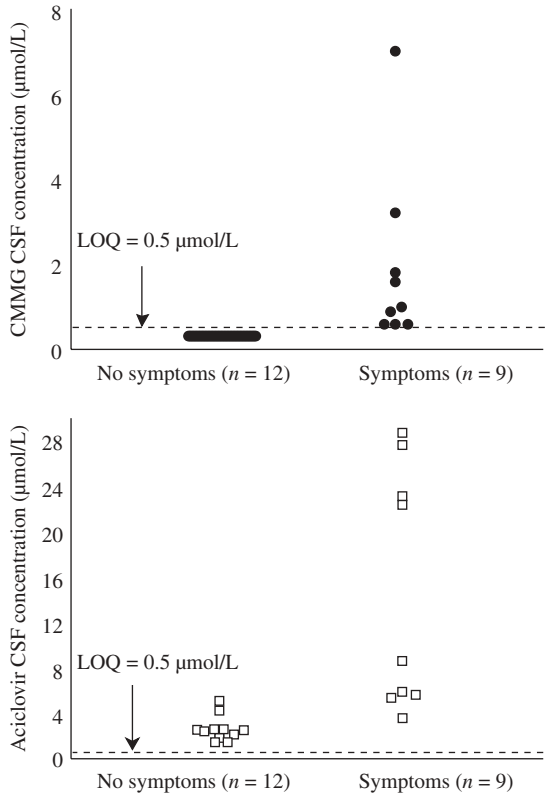



**Figure 2.** CSF concentrations of 9-carboxymethoxymethylguanine (CMMG) and aciclovir (ACV) from 12 subjects without and 9 subjects with neuropsychiatric symptoms. CMMG was detectable in the CSF from symptomatic patients. The 12 subjects without neuropsychiatric symptoms had CMMG CSF concentrations below the limit of quantification (LOQ) = 0.5 μmol/L. To convert values for ACV concentrations to μg/mL, divide by 4.44. To convert values for CMMG concentrations to μg/mL, divide by 4.18. Open squares = aciclovir concentrations; filled circles = CMMG concentrations. ACV = aciclovir; CMMG = 9-carboxymethoxymethylguanine.

## Discussion

In a previous study we found that patients who developed neuropsychiatric side effects during treatment with aciclovir or valaciclovir had high concentrations of the aciclovir main metabolite CMMG in serum.[14] This finding prompted the hypothesis that CMMG may induce neuropsychiatric side effects. To be valid, this hypothesis requires that CMMG be shown to exist in the CNS. We have shown here the presence of CMMG in the CSF, but only in subjects with neuropsychiatric signs and symptoms. We conclude that CMMG enters the CSF.

A methodological issue concerns the pharmacokinetics of aciclovir and CMMG and the data used in this study. The basic assumption made here is that the CSF concentrations of aciclovir and CMMG are directly dependent on their concentrations in blood. The samples were collected after a longer time interval from the administered dose in some subjects with symptoms than in those without symptoms. Also, since almost all subjects with symptoms had renal insufficiency, the half-lives of both aciclovir and CMMG were prolonged and the concentrations of both compounds were elevated. It may be argued that the difference in CMMG CSF concentrations between subjects with or without symptoms is due to the systematic difference in sampling times and half-lives. However, the asymptomatic subjects were at steady state, and, as shown in an earlier publication, the aciclovir concentrations in the CSF in the 2 and 8 h samples did not differ.[15] Furthermore, the subjects with signs and symptoms were sampled at varying times after the last dose and in some cases the concentrations of CMMG and aciclovir may have been declining. We therefore conclude that the increased CSF concentrations of aciclovir and CMMG are due to high serum concentrations of the compounds.

Another issue concerns the CSF sample that was analysed several months after it was taken. However, preliminary studies at our laboratory show that aciclovir and CMMG are very stable compounds and that the concentrations in samples stored for more than 3 years vary <10%.

It has previously been found that the CSF aciclovir concentrations were constant between valaciclovir doses whereas the serum aciclovir concentrations varied considerably.[15] This finding suggests that CSF might function as a reservoir with a slow equilibrium



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capacity. Thus, at steady state, the CSF aciclovir and CMMG concentrations are probably essentially stable, and it can be assumed that the time relation between intake of drug and CSF sampling has little influence on the CSF concentration of aciclovir and CMMG in the present study. In general, diffusion of drugs into and out of the CSF is slow, as has been shown for several drugs.[19]

The mechanism for the presence of CMMG in CSF is not yet known. ADH and ALDH probably form CMMG from aciclovir (Figure 1).[11–13] ADH and ALDH are abundant in the liver with only minor contributions from the CNS. It is reasonable to assume that the amount of enzyme in the CNS is too small to yield the CMMG concentrations observed in this study and that CMMG is transferred to the CSF. The high serum concentrations of CMMG present in subjects with neuropsychiatric side effects may predispose to CSF transfer. We therefore suggest that the passage of CMMG across the blood–CSF barrier is the main reason for the presence of CMMG in CSF. However, this needs to be confirmed in other studies.

The finding that increased concentrations of aciclovir are found in the CSF in symptomatic subjects is also of importance. Four subjects with symptoms had CSF aciclovir concentrations equivalent to those found in serum. Hence, we cannot exclude the possibility that aciclovir contributed to the adverse reactions. On the other hand, two asymptomatic subjects (4 and 11) had CSF aciclovir concentrations similar to those in the symptomatic subjects. If aciclovir had been responsible for the neuropsychiatric symptoms, these subjects would have been expected to have had side effects too.

It is interesting that oral aciclovir treatment using 2000 mg daily gave rise to neuropsychiatric symptoms in a patient with normal renal function.[7] This patient had a CSF aciclovir concentration similar to those in some of our asymptomatic patients but also had measurable CSF concentrations of CMMG. Thus, it seems that acute or chronic renal failure is not always a prerequisite for the development of neuropsychiatric side effects during aciclovir therapy.

We conclude that there are two sets of evidence suggesting that the neuropsychiatric side effects of aciclovir are related to CMMG: (i) high serum concentrations of CMMG are found in aciclovir- or valaciclovir-treated subjects with neuropsychiatric side effects and (ii) CMMG is detectable only in the CSF of subjects with neuropsychiatric signs and symptoms.

In clinical practice, measurement of CMMG in CSF and/or serum might be an important tool for the diagnosis of aciclovir-induced neuropsychiatric symptoms, especially in aciclovir- or valaciclovir-treated patients with neurological symptoms of unclear origin.

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## Transparency declarations

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## References

**1.** Vartian C, Shlaes D. Intravenous acyclovir and neurologic effects. *Ann Intern Med* 1983; **99**: 568.

**2.** Tilson H. Monitoring the safety of antivirals. *Am J Med* 1988; **85** Suppl 2A: 116–22.

**3.** Linssen-Schuurmans CD, van Kan EJ, Feith GW *et al.* Neurotoxicity caused by valacyclovir in a patient on hemodialysis. *Ther Drug Monit* 1998; **30**: 385–6.

**4.** Peyriere H, Branger B, Bengler C *et al.* Toxicite neurologique induite par le Zelitrex (valaciclovir) chez trois patients insuffisants renaux. Surdosage lie a l'amelioration de la biodisponibilite du produit? *Rev Med Interne* 2000; **28**: 297–303.

**5.** Wade J, Meyers J. Neurologic symptoms associated with parenteral acyclovir treatment after marrow transplantation. *Ann Intern Med* 1983; **98**: 921–5.

**6.** Itoh M, Hayakawa K. A girl with transient psychological disturbance caused by orally administered acyclovir—differentiation between acyclovir neurotoxicity and herpes simplex encephalitis. *No To Hattatsu* 1998; **78**: 328–33.

**7.** Vander T, Medvedovsky M, Herishanu Y. Encephalopathy induced by oral acyclovir in a patient with normal renal function. *J Infect* 2003; **46**: 286.

**8.** Feldman S, Rodman J, Gregory B. Excessive serum concentrations of acyclovir and neurotoxicity. *J Infect Dis* 1988; **157**: 385–8.

**9.** de Knegt RJ, van der Pijl H, van Es LA. Acyclovir-associated encephalopathy, lack of relationship between acyclovir levels and symptoms. *Nephrol Dial Transplant* 1995; **10**: 1775–7.

**10.** Haefeli WE, Schoenenberger RA, Weiss P *et al.* Acyclovir-induced neurotoxicity: concentration-side effect relationship in acyclovir overdose. *Am J Med* 1993; **94**: 212–5.

**11.** de Miranda P, Good SS. Metabolic fate and pharmacokinetics of the acyclovir prodrug valaciclovir in cynomolgus monkeys. *Drug Metab Dispos* 1994; **22**: 55–9.

**12.** de Miranda P, Burnette TC. Species differences in the metabolism and disposition of antiviral nucleoside analogues. *Antiviral Chem Chemother* 1992; **18**: 1–8

**13.** de Miranda P, Good S, Laskin OL *et al.* Disposition of intravenous radioactive acyclovir. *Clin Pharmacol Ther* 1981; **30**: 662–72

**14.** Helldén A, Odar-Cederlöf I, Diener P *et al.* High serum concentrations of the acyclovir main metabolite 9-carboxymethoxymethylguanine in renal failure patients with acyclovir-related neuropsychiatric side effects: an observational study. *Nephrol Dial Transplant* 2003; **18**: 1135–41.

**15.** Lycke J, Malmeström C, Ståhle L. Acyclovir levels in serum and cerebrospinal fluid after oral administration of valacyclovir. *Antimicrob Agents Chemother* 2003; **47**: 2438–41.

**16.** Bech E, Lycke J, Gadeberg P *et al.* A randomized, double-blind, placebo-controlled MRI study of anti-herpes virus therapy in MS. *Neurology* 2002; **58**: 31–6.

**17.** Svensson JO, Barkholt L, Säwe J. Determination of acyclovir and its metabolite 9-carboxymethoxymethylguanine in serum and urine using solid-phase extraction and high-performance liquid chromatography. *J Chromatogr B Biomed Sci Appl* 1997; **690**: 363–6.

**18.** Cockcroft DW, Gault MH. Prediction of creatinine clearance from serum creatinine. *Nephron* 1976; **16**: 31–41.

**19.** Strazielle N, Ghersi-Egea JF. Factors affecting delivery of antiviral drugs to the brain. *Rev Med Virol* 2005; **15**: 105–33.



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